Syed Jawad Ali

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Technical Skills

Languages: Python, JavaScript, TypeScript
Frameworks: Django, FastAPI, Next.js
Databases: PostgreSQL, MySQL, SQLite, pgvector
Generative AI & LLM: Retrieval-Augmented Generation (RAG), LLM Application Development, Prompt Engineering, LangChain, LangGraph, Agentic and Multi-Agent Systems
Vector Databases: ChromaDB, PostgreSQL Vector Store
ML & AI Libraries: TensorFlow, Keras, scikit-learn, Pandas, NumPy
Tools & Platforms: Docker, Git, Redis, Tailwind CSS, REST APIs, Agile (Jira)

Experience

Joyn Digital, LiftUpServices Dec 2024 – Present
Freelance Python Developer (Project-Based — Django, FastAPI, GenAI – LLM/RAG — Remote) *Islamabad, Pakistan*

- Designed and developed scalable backend systems using FastAPI and Django REST Framework, serving AI-powered applications.
- Built production-grade RAG (Retrieval-Augmented Generation) systems for intelligent document querying with citation support. Developed agentic AI pipelines integrating LLMs, enabling automated reasoning, task planning, and multi-step decision workflows.
- Integrated LLMs (BERT, Transformers, OpenAI APIs) with backend services to deliver real-time AI features. Implemented vector search systems (embeddings, similarity search) using PostgreSQL and optimized retrieval performance. Collaborated with cross-functional teams to deploy and scale AI services in production environments.

BUIC Research Lab April 2018 – May 2020
Machine Learning Engineer *Islamabad, Pakistan*

- Hands-on experience with machine learning libraries including Pandas, NumPy, Matplotlib, scikit-learn, Keras, and TensorFlow.
- Applied Machine Learning techniques including Supervised and Unsupervised algorithms for Classification, Clustering, and Dimensionality Reduction, as well as NLP using the Bidirectional Encoder Representations from Transformers (BERT) model and CNNs.

NextIn Technologies August 2016 – Dec 2018
Dot Net Developer *Islamabad, Pakistan*

- Worked as ASP.NET Web Form and MVC developer.
- Transformed business logic into scalable backend solutions and collaborated with cross-functional teams to design, implement, and optimize systems.

Projects

Intelligent RAG SaaS Platform (AI Knowledge Assistant) | *Django and Next.js*

- Developed a production-level RAG-based SaaS system enabling companies to query internal documents (PDF, CSV, Docs) using natural language.
- Implemented hybrid search (vector + keyword) with PostgreSQL + pgvector for high-accuracy retrieval. Designed context-aware response generation with citations, improving answer reliability.
- Built features including role-based access, conversation memory, and admin dashboard. Enhanced system performance using re-ranking, query rewriting, and confidence scoring.

Multi-Agent AI System with Decision Intelligence (RAG + RL) | *Django and Next.js*

- Designed a multi-agent AI system with Planner, Retrieval, and Answer agents for complex task execution. Implemented an RL-based decision module to dynamically choose between retrieving more data, answering, or asking clarification.
- Defined state-action-reward framework to optimize accuracy, reduce hallucination, and minimize API cost. Built adaptive pipelines that improve response quality through intelligent decision-making.

- Implemented role-based access control, conversation memory, and admin dashboard features, while improving system performance through re-ranking, query rewriting, and confidence scoring.

Intelligent LMS with Agentic RAG System (AI-Powered Learning Platform) | *Django and Next.js*

- Developed a full-featured Learning Management System (LMS) with modular architecture covering academic management, attendance, exams, assignments, communication, notifications, and reporting.
- Designed and implemented an Agentic LLM-based RAG system to enable intelligent question answering over course materials and institutional data. Integrated PostgreSQL with vector storage (pgvector) and ChromaDB for efficient embedding storage and semantic retrieval.
- Built a hybrid retrieval pipeline (vector + structured data) to enhance answer accuracy and contextual relevance. Implemented multi-agent architecture (planner, retrieval, and response agents) to handle complex user queries and workflows. Enabled context-aware AI interactions for querying academic data using natural language.

AI Resume Screening System | *Django and Next.js*

- Developed an AI-powered system to analyze and rank resumes based on job descriptions.
- Applied NLP techniques and embeddings to match candidate skills with job requirements. Generated explainable scoring and insights for HR decision-making.

Business Digitalization System | *Django*

- A Business Digitalization and Enterprise Resource Management System developed using Django, React.js, and deployed on Amazon EC2 cloud infrastructure.
- A Machine Learning trained model is integrated into the backend for predictive analysis and decision-making.

Human Resource Management System | *Django, FastAPI and Next.js*

- A robust platform designed to efficiently manage employee records, attendance, and departmental data. The backend is built using Django REST Framework and FastAPI to ensure scalability and performance. The frontend is developed with Next.js, delivering a fast, responsive, and modern user experience.
- Integrated Machine Learning models are used to predict employee salaries based on experience and education, and to evaluate performance using attendance and other key factors.

Scalable E-commerce Platform (Product, Order and Inventory System) | *Django, Next.js and PostgreSQL*

- Developed a full-featured e-commerce platform supporting product management, categories, variants (size, color, material), and dynamic product tagging for advanced filtering.
- Designed and implemented inventory and stock management system with real-time tracking, reorder levels, and automated SKU generation for product variants.
- Built order management and checkout system including cart, order processing, payment handling, and delivery tracking with multiple payment methods. Implemented sales and promotion modules with discount logic, dynamic pricing, and automated calculation of final prices.

Blogging Platform with Scalable Content Management System | *Django, Next.js and PostgreSQL*

- Developed a full-featured blogging platform with modular architecture supporting categories, tags, posts, media management, comments, newsletters, and campaign systems. Designed and implemented advanced content models with SEO optimization, hierarchical categorization, and rich text editing to improve content organization and discoverability.
- Built a robust comment system with moderation workflows, nested replies, rate limiting, and role-based access control to ensure secure and scalable user engagement. Integrated media library and newsletter campaign management with subscriber segmentation and scheduling for efficient content distribution.

Education

Bahria University — Islamabad, Pakistan
Master of Science in Computer Science (CGPA: 3.41) — *2020*

- **Relevant Coursework:** Data Science, Machine Learning, Deep Learning, and AI-based Thesis
- **Relevant Research:** Selection of Web Services using Artificial Intelligence

Federal Urdu University of Science and Arts — Islamabad, Pakistan
Bachelor of Science in Computer Science (CGPA: 3.01) — *2016*

- **Relevant Coursework:** Data Structures and Algorithms (C++), Programming Fundamentals (C++, Python), Web and Mobile Applications